

TriVoKo

Phase 1 - Catalogue API

Time	Start date	Finish date	Status
1 week			Not started

Goal: Products exist in the database and anyone can browse them through the API: categories, filters, sorting, paging and product details.

Real-life example: The shelves are filled and every product has a price tag and a label. Customers can walk the aisles, but the billing counter is not open yet.

Step checklist

★ = you type the key logic (Claude explains first, then reviews). "You" = only you can do it.

#	Step	Who	Done
1	Flyway V1: categories (tree with parent), sellers (shop name, slug, city, status), products (status DRAFT / PENDING / ACTIVE / REJECTED / BLOCKED), product_variants (size, colour, price, MRP, stock, version), product_images, price_history	Claude	☐
2	Flyway V2 seed data from the sample-data plan: 10 categories, 8 sellers, 120 products with variants and photos	Claude	☐
3	Entities and repositories in modules catalog and seller; money as BigDecimal / DECIMAL(12, 2)	Claude	☐
4	ArchUnit test for the module rule (a module may call another module's service, never its repository) - runs in CI from now on	Claude	☐
5	DTO records + mapper; shared PageResponse	Claude	☐
6	GET /api/products with filters (category incl. sub-categories, brand, min/max price, in stock), sort (price, newest) and paging (max 48)	Claude	☐
7	GET /api/products/{slug} (variants, images, seller), GET /api/categories (tree), GET /api/sellers/{slug} (shop page)	Claude	☐
8	Price-change service method that always writes price_history (used by price alerts in Phase 9)	Claude	☐
9	Create a free Cloudinary account and put the 3 keys in .env	You	☐
10	Cloudinary signed upload: POST /api/uploads/signature (images only, max 5 MB) - tried with one real photo	Claude	☐
11	Tests: repository tests, service rules, every endpoint (Testcontainers MySQL)	Claude	☐
12	Postman collection with checks, run with newman	Claude	☐

Output of this phase

- Catalogue REST API in Swagger
- 120 sample products in MySQL

- Postman collection
- ArchUnit module rule in CI

Done when (check before moving on)

Check	Passed
GET /api/products?category=phones&sort=price returns the correct pages	☐
All Postman checks pass	☐
CI green	☐

Explain it back (in your own words)

Question	Done
Why does stock live on the variant and not on the product?	☐
What is a Flyway migration and why do we never change the database by hand?	☐
How do filters become one SQL query (Specifications)?	☐

Finish routine

Check	Done
All tests pass and CI is green	☐
Committed and pushed	☐
Done-when checks shown working in the browser	☐
Session summary + SESSION_LOG written	☐

Skills you practise: JPA entities, relationships, Flyway, Specifications, paging, DTOs, testing with Testcontainers

Where your work goes: `backend\src\main\java\com\trivoko\catalog, ...\seller, db\migration, postman\`

My notes

